

```

srcn_augueb_rented_houses_4 (1).ino
8
9
10 #include <elapsedMillis.h> // reqd to have timing
11 elapsedMillis timeElapsed1; // reqd to have relay1 timing
12 elapsedMillis timeElapsed2; // reqd to have relay2 timing
13 elapsedMillis timeElapsed3; // reqd to have relay3 timing
14 elapsedMillis timeElapsed4; // reqd to have relay4 timing
15 elapsedMillis timeElapsed5; // reqd to have flasher
16
17 int sw1 = 2;
18 int sw2 = 3;
19 int sw3 = 4;
20 int sw4 = 5;
21
22 int relay1 = 8;
23 int relay2 = 9;
24 int relay3 = 10;
25 int relay4 = 11;
26 int pinLED = 13;
27
28 int M1 = LOW;
29 int M2 = LOW;
30 int M3 = LOW;
31 int M4 = LOW;
32
33 int sw1State;
34 int sw2State;
35 int sw3State;
36 int sw4State;
37
38 int T1 = LOW;

```

Fig. 3: Code snippet with pin definitions for inputs and outputs

```

39 digitalWrite(relay2, T2);
40
41
42 if (M3 == HIGH)
43 {
44     if (timeElapsed3 <= 62000)
45     {
46         T3 = LOW;
47     }
48     else if (timeElapsed3 > 62000)
49     {
50         T3 = HIGH;
51     }
52 }
53 else if (M3 == LOW)
54 {
55     T3 = LOW;
56     timeElapsed3 = 0;
57 }
58 digitalWrite(relay3, T3);
59
60
61 if (M4 == HIGH)
62 {
63     if (timeElapsed4 <= 63000)
64     {
65         T4 = LOW;
66     }
67     else if (timeElapsed4 > 63000)

```

Fig. 4: Code snippet time lapse logic for each house



Fig. 5: Water valve and water motor setup for the houses

highest priority and starts the motor after a 60-second delay. Subsequently, all other houses are prevented from using the motor.

After house number 10 has completed its cycle, the motor can be activated again after a 61-second delay if house number 11 requires it. This pattern continues for the other two houses with decreasing priority, with delays of 62 and 63 seconds, respectively.

Whenever a house turns on the motor switch, its Arduino input pin (D2 through D5) goes low, and the corresponding output pin (D8 through D11) becomes high after the preset delay. This delay is based on the ascending order of input numbers, and the relay is energised through the ULN2003 driver. As a result, the motor runs on power from the respective house,

and the program prevents other house relays from activating. Even if other houses turn on their motor switches, the motor won't change its supply source, until the current user house turns off their motor switch. If two or more houses turn on their switches simultaneously, the motor starts in priority order, as explained earlier.

The built-in LED of the Nano is programmed to flash (0.5 seconds on, 0.5 seconds off) continuously to indicate that the program loop is active. If the flashing stops, it indicates a hung state of the Arduino Nano, which can be resolved by resetting the Nano by pressing the reset button or by power-cycling the control supply.

Software

The software is written in the Arduino programming language using Arduino IDE version 1.8.7. Before uploading the 'rentedhouses4.ino' sketch/program to the Arduino Nano, ensure that the 'elapsedMillis' library function is available in the Arduino IDE. Upload the sketch to the board after selecting the appropriate board